

Replication-level Review

2026-03-22

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

rep_binary <- read.table("~/Desktop/rep_binary.dat", quote="\"", comment.char="")
colnames(rep_binary) <- c("categories", "group_prob", "rsq_prod", "sample_size",
                          "loading", "n_items", "model_type", "param.id", "est",
                          "se", "true", "bias", "rel.bias", "squared_bias", "pval",
                          "sig", "ci.cov", "ci.zero", "joint.stat", "joint.p",
                          "no_cvg_check", "psr.max", "neff.min", "ci.low", "ci.up",
                          "rep", "seed")

summarise_condition <- function(data, loading_val = 0.5, group_prob_val = 2) {

  dat <- data %>%
    filter(
      loading == loading_val,
      group_prob == group_prob_val,
      rsq_prod == 0.03
    ) %>%
    mutate(
      crashed = is.na(est),
      exclude_psr = psr.max > 1.05,
      exclude_neff = neff.min < 100,
      exclude_any = exclude_psr | exclude_neff
    )

  total_reps <- length(unique(dat$rep))
  cat("Found", total_reps, "replications\n")
  cat("Found", nrow(dat), "rows of results\n")

  for (n in sort(unique(dat$sample_size))) {

    cat("\n#####\n")
  }
}
```

```

cat("N =", n, "\n")
cat("#####\n")

d <- dat %>% filter(sample_size == n)

attempted <- length(unique(d$rep))
crashed <- sum(tapply(d$crashed, d$rep, any))
successful <- attempted - crashed

cat(sprintf("Crashed reps: %d (%.1f%%)\n", crashed, 100 * crashed / attempted))
cat(sprintf("Successful reps: %d (%.1f%%)\n", successful, 100 * successful / attempted))

# -----
# ALL successful reps
# -----
cat("\n-- Parameter Recovery (all successful reps) --\n")

d_all <- d %>% filter(!crashed)

emp_sd_df <- d_all %>%
  group_by(param.id) %>%
  summarise(emp_sd = sd(est, na.rm = TRUE), .groups = "drop")

summ_all <- d_all %>%
  group_by(param.id) %>%
  summarise(
    true = mean(true, na.rm = TRUE),
    est = mean(est, na.rm = TRUE),
    avg_post_sd = mean(se, na.rm = TRUE),
    pct_sig = mean(sig, na.rm = TRUE),
    .groups = "drop")

summ_all <- summ_all %>%
  left_join(emp_sd_df, by = "param.id") %>%
  mutate(
    bias_sd = (est - true) / emp_sd,
    z = (est - true) / avg_post_sd)

p1 <- summ_all %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

print(as.data.frame(p1))

# -----
# FILTERED reps
# -----
cat("\n-- Parameter Recovery (excluding high PSR or low N_eff) --\n")

d_filt <- d_all %>% filter(!exclude_any)

emp_sd_filt <- d_filt %>%
  group_by(param.id) %>%
  summarise(emp_sd = sd(est, na.rm = TRUE), .groups = "drop")

```

```

summ_filt <- d_filt %>%
  group_by(param.id) %>%
  summarise(
    true = mean(true, na.rm = TRUE),
    est = mean(est, na.rm = TRUE),
    avg_post_sd = mean(se, na.rm = TRUE),
    pct_sig = mean(sig, na.rm = TRUE),
    .groups = "drop")

summ_filt <- summ_filt %>%
  left_join(emp_sd_filt, by = "param.id") %>%
  mutate(
    bias_sd = (est - true) / emp_sd,
    z = (est - true) / avg_post_sd)

p2<- summ_filt %>%
  mutate(across(where(is.numeric), ~ round(.x, 4)))

print(as.data.frame(p2))

# -----
# Convergence summary
# -----
cat("\n-- Convergence / N_eff Summary (all successful reps) --\n")

conv <- d_all %>%
  summarise(
    psr_min = min(psr.max, na.rm = TRUE),
    psr_q1 = quantile(psr.max, .25, na.rm = TRUE),
    psr_med = median(psr.max, na.rm = TRUE),
    psr_mean = mean(psr.max, na.rm = TRUE),
    psr_q3 = quantile(psr.max, .75, na.rm = TRUE),
    psr_max = max(psr.max, na.rm = TRUE),

    neff_min = min(neff.min, na.rm = TRUE),
    neff_q1 = quantile(neff.min, .25, na.rm = TRUE),
    neff_med = median(neff.min, na.rm = TRUE),
    neff_mean = mean(neff.min, na.rm = TRUE),
    neff_q3 = quantile(neff.min, .75, na.rm = TRUE),
    neff_max = max(neff.min, na.rm = TRUE)
  )

print(round(conv, 3))
}
}

```

```
# LOADING = .50 SPLIT = 50/50
summarise_condition(rep_binary, loading_val = 0.5, group_prob_val = 1)
```

```
## Found 200 replications
## Found 30000 rows of results
##
## #####
## N = 200
## #####
## Crashed reps: 87 (43.5%)
## Successful reps: 113 (56.5%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0091    0.1380  0.0431 0.1421 -0.0107 -0.0110
## 2  beta_G2 0.0000 0.0127    0.1496  0.0566 0.1531  0.0832  0.0852
## 3  beta_X  0.3606 0.2882    0.1252  0.6810 0.1476 -0.4901 -0.5780
## 4 beta_XG2 0.0755 0.0627    0.1644  0.0404 0.1617 -0.0787 -0.0774
## 5  res.var 0.8400 0.7611    0.2671  0.6217 0.2151 -0.3668 -0.2953
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0084    0.1416  0.0519 0.1480 -0.0056 -0.0058
## 2  beta_G2 0.0000 0.0095    0.1552  0.0593 0.1613  0.0589  0.0612
## 3  beta_X  0.3606 0.3022    0.1279  0.6827 0.1543 -0.3784 -0.4566
## 4 beta_XG2 0.0755 0.0657    0.1720  0.0494 0.1754 -0.0556 -0.0567
## 5  res.var 0.8400 0.7936    0.2765  0.6556 0.2202 -0.2109 -0.1680
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004  1.008   1.506  1.015 12.646   1.004 181.807 283.688
##   neff_mean neff_q3 neff_max
## 1   307.332 432.201 822.616
##
## #####
## N = 300
## #####
## Crashed reps: 10 (5.0%)
## Successful reps: 190 (95.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0206    0.1130  0.0529 0.1155 -0.1128 -0.1153
## 2  beta_G2 0.0000 0.0206    0.1207  0.0412 0.1232  0.1672  0.1706
## 3  beta_X  0.3606 0.3001    0.1042  0.8639 0.1274 -0.4742 -0.5801
## 4 beta_XG2 0.0755 0.0529    0.1337  0.0630 0.1273 -0.1772 -0.1687
## 5  res.var 0.8400 0.7554    0.2137  0.6630 0.1886 -0.4489 -0.3960
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0217    0.1137  0.0548 0.1162 -0.1215 -0.1241
## 2  beta_G2 0.0000 0.0166    0.1221  0.0415 0.1238  0.1342  0.1361
## 3  beta_X  0.3606 0.3087    0.1010  0.8716 0.1244 -0.4169 -0.5135
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## 4 beta_XG2  0.0755  0.0549      0.1344  0.0623  0.1299 -0.1581 -0.1528
## 5 res.var  0.8400  0.7674      0.2166  0.6667  0.1867 -0.3887 -0.3350
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.006    1.703  1.011  19.976    1.002 225.313  345.835
##   neff_mean neff_q3 neff_max
## 1   341.355 454.792  899.498
##
## #####
## N = 400
## #####
## Crashed reps: 1 (0.5%)
## Successful reps: 199 (99.5%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id  true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0141    0.0986  0.0525  0.1034 -0.0629 -0.0660
## 2   beta_G2 0.0000  0.0100    0.1059  0.0300  0.1023  0.0980  0.0946
## 3   beta_X  0.3606  0.3104    0.0886  0.9575  0.1078 -0.4650 -0.5659
## 4  beta_XG2 0.0755  0.0614    0.1164  0.0817  0.1202 -0.1166 -0.1204
## 5  res.var  0.8400  0.7774    0.1877  0.6664  0.1640 -0.3817 -0.3335
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id  true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0132    0.0987  0.0527  0.1035 -0.0546 -0.0572
## 2   beta_G2 0.0000  0.0096    0.1061  0.0298  0.1023  0.0941  0.0908
## 3   beta_X  0.3606  0.3123    0.0874  0.9617  0.1061 -0.4550 -0.5522
## 4  beta_XG2 0.0755  0.0618    0.1165  0.0833  0.1210 -0.1125 -0.1168
## 5  res.var  0.8400  0.7804    0.1882  0.6667  0.1631 -0.3653 -0.3165
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.003   1.006    1.174  1.009  17.126    1.003 308.654  394.397
##   neff_mean neff_q3 neff_max
## 1         405 502.728  973.898
##
## #####
## N = 500
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id  true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0067    0.0874  0.0450  0.0893  0.0099  0.0101
## 2   beta_G2 0.0000  0.0061    0.0934  0.0550  0.0966  0.0634  0.0656
## 3   beta_X  0.3606  0.3004    0.0773  0.9758  0.0982 -0.6125 -0.7788
## 4  beta_XG2 0.0755  0.0675    0.1026  0.1042  0.1036 -0.0772 -0.0780
## 5  res.var  0.8400  0.7592    0.1660  0.6667  0.1464 -0.5518 -0.4866
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id  true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0064    0.0875  0.0453  0.0895  0.0129  0.0132

```

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## 2  beta_G2  0.0000  0.0056      0.0934  0.0546 0.0967  0.0581  0.0601
## 3  beta_X   0.3606  0.3011      0.0768  0.9773 0.0976 -0.6090 -0.7742
## 4  beta_XG2 0.0755  0.0674      0.1026  0.1024 0.1034 -0.0780 -0.0786
## 5  res.var  0.8400  0.7601      0.1661  0.6667 0.1464 -0.5458 -0.4810
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.003   1.005    1.092  1.008  18.407    1.002 310.177 395.254
##   neff_mean neff_q3 neff_max
## 1    401.03 481.216 783.494
##
## #####
## N = 1000
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0122    0.0620  0.0608 0.0638 -0.0727 -0.0748
## 2   beta_G2 0.0000  0.0123    0.0665  0.0633 0.0690  0.1781  0.1847
## 3   beta_X  0.3606  0.3093    0.0549  0.9992 0.0835 -0.6142 -0.9340
## 4  beta_XG2 0.0755  0.0619    0.0730  0.1567 0.0783 -0.1728 -0.1854
## 5  res.var  0.8400  0.7807    0.1188  0.6667 0.1182 -0.5016 -0.4992
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0124    0.0620  0.0610 0.0637 -0.0758 -0.0778
## 2   beta_G2 0.0000  0.0123    0.0665  0.0635 0.0691  0.1774  0.1842
## 3   beta_X  0.3606  0.3095    0.0546  1.0000 0.0830 -0.6148 -0.9354
## 4  beta_XG2 0.0755  0.0621    0.0729  0.1571 0.0783 -0.1708 -0.1835
## 5  res.var  0.8400  0.7806    0.1187  0.6667 0.1182 -0.5027 -0.5006
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.003   1.004    1.048  1.007  17.775    1.003 385.368 449.128
##   neff_mean neff_q3 neff_max
## 1    455.897 515.248 796.201

```

```
# LOADING = .80 SPLIT = 50/50
summarise_condition(rep_binary, loading_val = 0.8, group_prob_val = 1)
```

```
## Found 200 replications
## Found 30000 rows of results
##
## #####
## N = 200
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0109      0.1023  0.0500 0.1069 -0.0313 -0.0328
## 2  beta_G2 0.0000 0.0079      0.1314  0.0592 0.1366  0.0581  0.0604
## 3  beta_X  0.3606 0.3329      0.0996  0.9208 0.1132 -0.2446 -0.2778
## 4 beta_XG2 0.0755 0.0703      0.1356  0.0883 0.1375 -0.0376 -0.0381
## 5  res.var 0.8400 0.7639      0.1272  0.6667 0.1380 -0.5516 -0.5985
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0115      0.1023  0.0498 0.1072 -0.0365 -0.0383
## 2  beta_G2 0.0000 0.0078      0.1314  0.0610 0.1379  0.0568  0.0596
## 3  beta_X  0.3606 0.3364      0.0975  0.9287 0.1093 -0.2210 -0.2475
## 4 beta_XG2 0.0755 0.0705      0.1348  0.0868 0.1378 -0.0357 -0.0365
## 5  res.var 0.8400 0.7644      0.1273  0.6667 0.1376 -0.5494 -0.5939
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004  1.007    2.059  1.012 63.229    1.001 197.155 275.336
##   neff_mean neff_q3 neff_max
## 1   298.969 405.257 740.874
##
## #####
## N = 300
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0093      0.0836  0.0458 0.0785 -0.0224 -0.0210
## 2  beta_G2 0.0000 0.0009      0.1075  0.0733 0.1102  0.0081  0.0083
## 3  beta_X  0.3606 0.3327      0.0797  0.9883 0.0871 -0.3199 -0.3498
## 4 beta_XG2 0.0755 0.0766      0.1097  0.1017 0.1058  0.0113  0.0109
## 5  res.var 0.8400 0.7728      0.1040  0.6667 0.1264 -0.5321 -0.6469
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0075 -0.0092      0.0836  0.0453 0.0782 -0.0207 -0.0194
## 2  beta_G2 0.0000 0.0012      0.1075  0.0739 0.1098  0.0108  0.0110
## 3  beta_X  0.3606 0.3330      0.0791  0.9899 0.0863 -0.3193 -0.3485
```

```

## 4 beta_XG2 0.0755 0.0769 0.1097 0.1024 0.1060 0.0138 0.0134
## 5 res.var 0.8400 0.7728 0.1040 0.6667 0.1264 -0.5319 -0.6470
##
## -- Convergence / N_eff Summary (all successful reps) --
## psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1 1.001 1.004 1.007 1.242 1.011 47.866 1.001 214.948 291.986
## neff_mean neff_q3 neff_max
## 1 317.786 418.384 706.64
##
## #####
## N = 400
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
## param.id true est avg_post_sd pct_sig emp_sd bias_sd z
## 1 beta_0 -0.0075 -0.0110 0.0722 0.0567 0.0764 -0.0451 -0.0478
## 2 beta_G2 0.0000 0.0091 0.0928 0.0608 0.0951 0.0956 0.0980
## 3 beta_X 0.3606 0.3374 0.0686 0.9975 0.0788 -0.2945 -0.3384
## 4 beta_XG2 0.0755 0.0611 0.0944 0.1083 0.0960 -0.1496 -0.1520
## 5 res.var 0.8400 0.7704 0.0898 0.6667 0.1183 -0.5879 -0.7749
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
## param.id true est avg_post_sd pct_sig emp_sd bias_sd z
## 1 beta_0 -0.0075 -0.0109 0.0722 0.0571 0.0764 -0.0435 -0.0461
## 2 beta_G2 0.0000 0.0089 0.0928 0.0613 0.0953 0.0931 0.0957
## 3 beta_X 0.3606 0.3375 0.0683 0.9983 0.0784 -0.2937 -0.3372
## 4 beta_XG2 0.0755 0.0614 0.0944 0.1092 0.0962 -0.1456 -0.1484
## 5 res.var 0.8400 0.7701 0.0897 0.6667 0.1178 -0.5935 -0.7793
##
## -- Convergence / N_eff Summary (all successful reps) --
## psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1 1.001 1.004 1.006 1.173 1.01 67.163 1.001 211.327 295.114
## neff_mean neff_q3 neff_max
## 1 315.49 413.409 620.786
##
## #####
## N = 500
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
## param.id true est avg_post_sd pct_sig emp_sd bias_sd z
## 1 beta_0 -0.0075 -0.0130 0.0645 0.0400 0.0612 -0.0892 -0.0846
## 2 beta_G2 0.0000 0.0087 0.0830 0.0383 0.0780 0.1112 0.1045
## 3 beta_X 0.3606 0.3337 0.0614 0.9992 0.0679 -0.3952 -0.4372
## 4 beta_XG2 0.0755 0.0659 0.0844 0.1083 0.0857 -0.1118 -0.1135
## 5 res.var 0.8400 0.7720 0.0804 0.6667 0.1106 -0.6151 -0.8466
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
## param.id true est avg_post_sd pct_sig emp_sd bias_sd z
## 1 beta_0 -0.0075 -0.0127 0.0645 0.0404 0.0613 -0.0843 -0.0801

```



```

## 2  beta_G2  0.0000  0.0084      0.0830  0.0387 0.0781  0.1078  0.1014
## 3  beta_X   0.3606  0.3340      0.0610  1.0000 0.0673 -0.3937 -0.4345
## 4  beta_XG2 0.0755  0.0660      0.0844  0.1094 0.0859 -0.1102 -0.1121
## 5  res.var  0.8400  0.7720      0.0804  0.6667 0.1107 -0.6148 -0.8465
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.006    1.18   1.01  69.687      1 218.248  300.542
##   neff_mean neff_q3 neff_max
## 1   325.929 433.755  628.671
##
## #####
## N = 1000
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0101    0.0457  0.0492 0.0444 -0.0573 -0.0556
## 2   beta_G2 0.0000 -0.0003    0.0589  0.0542 0.0571 -0.0060 -0.0058
## 3   beta_X  0.3606  0.3332    0.0430  1.0000 0.0563 -0.4863 -0.6365
## 4  beta_XG2 0.0755  0.0711    0.0596  0.2392 0.0617 -0.0700 -0.0724
## 5  res.var  0.8400  0.7779    0.0570  0.6667 0.1024 -0.6066 -1.0891
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0075 -0.0099    0.0457  0.0497 0.0444 -0.0522 -0.0506
## 2   beta_G2 0.0000 -0.0003    0.0589  0.0547 0.0573 -0.0054 -0.0052
## 3   beta_X  0.3606  0.3331    0.0430  1.0000 0.0563 -0.4878 -0.6382
## 4  beta_XG2 0.0755  0.0712    0.0597  0.2391 0.0618 -0.0689 -0.0714
## 5  res.var  0.8400  0.7782    0.0571  0.6667 0.1024 -0.6039 -1.0837
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.006    1.008   1.01   1.048   15.609 230.091  306.943
##   neff_mean neff_q3 neff_max
## 1   331.774 436.088  636.244

```

```
# LOADING = .50 SPLIT = 80/20
summarise_condition(rep_binary, loading_val = 0.5, group_prob_val = 2)
```

```
## Found 200 replications
## Found 30000 rows of results
##
## #####
## N = 200
## #####
## Crashed reps: 88 (44.0%)
## Successful reps: 112 (56.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0103    0.1203  0.0396 0.1207  0.0171  0.0171
## 2  beta_G2 0.0000  0.0148    0.2012  0.0558 0.2114  0.0701  0.0737
## 3  beta_X  0.3606  0.2948    0.1057  0.8822 0.1269 -0.5179 -0.6217
## 4 beta_XG2 0.1539  0.1334    0.2116  0.0890 0.2080 -0.0985 -0.0969
## 5  res.var 0.8400  0.7660    0.2661  0.6250 0.2134 -0.3470 -0.2782
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0084    0.1236  0.0388 0.1235  0.0314  0.0314
## 2  beta_G2 0.0000  0.0112    0.2114  0.0607 0.2270  0.0495  0.0531
## 3  beta_X  0.3606  0.3188    0.1056  0.8915 0.1254 -0.3331 -0.3957
## 4 beta_XG2 0.1539  0.1429    0.2217  0.0891 0.2229 -0.0491 -0.0494
## 5  res.var 0.8400  0.8097    0.2782  0.6628 0.2112 -0.1435 -0.1089
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004  1.007   1.859  1.016 17.592   1.003 179.459 318.798
##   neff_mean neff_q3 neff_max
## 1   315.626 447.163  923.137
##
## #####
## N = 300
## #####
## Crashed reps: 11 (5.5%)
## Successful reps: 189 (94.5%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0186    0.0987  0.0521 0.1007 -0.0627 -0.0639
## 2  beta_G2 0.0000  0.0264    0.1604  0.0513 0.1618  0.1629  0.1644
## 3  beta_X  0.3606  0.2973    0.0856  0.9529 0.1082 -0.5847 -0.7390
## 4 beta_XG2 0.1539  0.1236    0.1690  0.1152 0.1680 -0.1800 -0.1789
## 5  res.var 0.8400  0.7493    0.2134  0.6619 0.1848 -0.4910 -0.4251
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0197    0.0991  0.0524 0.1009 -0.0732 -0.0745
## 2  beta_G2 0.0000  0.0307    0.1623  0.0524 0.1633  0.1880  0.1892
## 3  beta_X  0.3606  0.3051    0.0820  0.9672 0.1037 -0.5350 -0.6769
```

```

## 4 beta_XG2  0.1539  0.1227      0.1695  0.1152 0.1725 -0.1805 -0.1837
## 5 res.var  0.8400  0.7618      0.2162  0.6667 0.1800 -0.4342 -0.3615
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.006    1.604  1.011 19.042    1.002 226.526 325.096
##   neff_mean neff_q3 neff_max
## 1   336.641 448.215 806.333
##
## #####
## N = 400
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0179    0.0858  0.0342 0.0854 -0.0658 -0.0655
## 2   beta_G2 0.0000  0.0235    0.1400  0.0542 0.1428  0.1644  0.1678
## 3   beta_X  0.3606  0.3054    0.0728  0.9900 0.0979 -0.5629 -0.7571
## 4 beta_XG2  0.1539  0.1354    0.1471  0.1433 0.1523 -0.1215 -0.1257
## 5 res.var  0.8400  0.7672    0.1861  0.6667 0.1746 -0.4168 -0.3909
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0179    0.0858  0.0337 0.0852 -0.0658 -0.0653
## 2   beta_G2 0.0000  0.0231    0.1403  0.0561 0.1442  0.1603  0.1648
## 3   beta_X  0.3606  0.3086    0.0709  0.9974 0.0947 -0.5488 -0.7332
## 4 beta_XG2  0.1539  0.1373    0.1465  0.1459 0.1527 -0.1087 -0.1134
## 5 res.var  0.8400  0.7712    0.1868  0.6667 0.1736 -0.3961 -0.3681
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.003   1.005    1.301  1.009 17.651    1.003 280.359 396.654
##   neff_mean neff_q3 neff_max
## 1   394.257 494.684 1113.809
##
## #####
## N = 500
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0155    0.0767  0.0633 0.0798 -0.0404 -0.0421
## 2   beta_G2 0.0000  0.0089    0.1252  0.0583 0.1282  0.0694  0.0710
## 3   beta_X  0.3606  0.3013    0.0638  0.9917 0.0910 -0.6514 -0.9296
## 4 beta_XG2  0.1539  0.1428    0.1304  0.2058 0.1452 -0.0761 -0.0848
## 5 res.var  0.8400  0.7720    0.1667  0.6667 0.1540 -0.4414 -0.4078
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0153    0.0767  0.0643 0.0800 -0.0368 -0.0384

```

```

## 2  beta_G2  0.0000  0.0088      0.1252  0.0592  0.1287  0.0683  0.0702
## 3  beta_X   0.3606  0.3027      0.0628  0.9949  0.0895 -0.6467 -0.9212
## 4  beta_XG2 0.1539  0.1435      0.1302  0.2090  0.1459 -0.0714 -0.0800
## 5  res.var  0.8400  0.7720      0.1666  0.6667  0.1538 -0.4424 -0.4083
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.003   1.005    1.143  1.008   18.27    1.002  305.27  397.435
##   neff_mean neff_q3 neff_max
## 1   408.015 500.218  770.173
##
## #####
## N = 1000
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id  true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0126    0.0543  0.0658 0.0587 -0.0051 -0.0056
## 2   beta_G2 0.0000  0.0177    0.0889  0.0875 0.0975  0.1815  0.1991
## 3   beta_X  0.3606  0.3091    0.0453  0.9983 0.0794 -0.6480 -1.1369
## 4  beta_XG2 0.1539  0.1336    0.0924  0.3108 0.0988 -0.2046 -0.2188
## 5  res.var  0.8400  0.7853    0.1192  0.6667 0.1234 -0.4433 -0.4591
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id  true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0127    0.0543  0.0662 0.0588 -0.0067 -0.0072
## 2   beta_G2 0.0000  0.0182    0.0889  0.0879 0.0974  0.1866  0.2046
## 3   beta_X  0.3606  0.3096    0.0447  1.0000 0.0783 -0.6509 -1.1403
## 4  beta_XG2 0.1539  0.1338    0.0922  0.3107 0.0989 -0.2025 -0.2170
## 5  res.var  0.8400  0.7852    0.1191  0.6667 0.1234 -0.4438 -0.4598
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.003   1.004    1.107  1.007  21.512    1.002  383.93  451.875
##   neff_mean neff_q3 neff_max
## 1   458.332 524.306  990.488

```

```
# LOADING = .80 SPLIT = 80/20
summarise_condition(rep_binary, loading_val = 0.8, group_prob_val = 2)
```

```
## Found 200 replications
## Found 30000 rows of results
##
## #####
## N = 200
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0112    0.0837  0.0392 0.0807  0.0138  0.0133
## 2  beta_G2 0.0000  0.0087    0.1736  0.0375 0.1681  0.0519  0.0503
## 3  beta_X  0.3606  0.3376    0.0800  0.9817 0.0885 -0.2597 -0.2874
## 4 beta_XG2 0.1539  0.1303    0.1706  0.1158 0.1627 -0.1449 -0.1382
## 5  res.var 0.8400  0.7631    0.1265  0.6667 0.1394 -0.5517 -0.6081
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0115    0.0837  0.0402 0.0813  0.0095  0.0092
## 2  beta_G2 0.0000  0.0083    0.1737  0.0385 0.1685  0.0491  0.0476
## 3  beta_X  0.3606  0.3398    0.0776  0.9889 0.0844 -0.2459 -0.2673
## 4 beta_XG2 0.1539  0.1313    0.1701  0.1188 0.1640 -0.1377 -0.1328
## 5  res.var 0.8400  0.7640    0.1266  0.6667 0.1393 -0.5454 -0.5999
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.005  1.007    1.94  1.012  53.991    1.001 196.663  274.787
##   neff_mean neff_q3 neff_max
## 1   298.751 396.456  882.083
##
## #####
## N = 300
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0142    0.0682  0.0600 0.0725 -0.0267 -0.0284
## 2  beta_G2 0.0000  0.0044    0.1415  0.0442 0.1434  0.0309  0.0313
## 3  beta_X  0.3606  0.3329    0.0631  0.9992 0.0745 -0.3708 -0.4380
## 4 beta_XG2 0.1539  0.1472    0.1381  0.2133 0.1498 -0.0446 -0.0483
## 5  res.var 0.8400  0.7657    0.1030  0.6667 0.1205 -0.6169 -0.7215
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1  beta_0 -0.0123 -0.0139    0.0681  0.0603 0.0725 -0.0218 -0.0232
## 2  beta_G2 0.0000  0.0042    0.1414  0.0444 0.1437  0.0293  0.0298
## 3  beta_X  0.3606  0.3333    0.0629  1.0000 0.0740 -0.3688 -0.4342
```

```

## 4 beta_XG2  0.1539  0.1471      0.1379  0.2127  0.1499 -0.0452 -0.0491
## 5 res.var  0.8400  0.7652      0.1029  0.6667  0.1203 -0.6212 -0.7262
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.007   1.157   1.01   60.34   1.001 216.558 295.876
##   neff_mean neff_q3 neff_max
## 1   314.453 407.056 649.316
##
## #####
## N = 400
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0171     0.0592  0.0667 0.0617 -0.0782 -0.0814
## 2   beta_G2 0.0000 -0.0016     0.1222  0.0633 0.1251 -0.0131 -0.0134
## 3   beta_X  0.3606  0.3323     0.0548  0.9992 0.0668 -0.4231 -0.5160
## 4 beta_XG2  0.1539  0.1404     0.1192  0.2358 0.1230 -0.1091 -0.1126
## 5 res.var  0.8400  0.7726     0.0901  0.6667 0.1173 -0.5744 -0.7480
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0170     0.0592  0.0670 0.0618 -0.0754 -0.0786
## 2   beta_G2 0.0000 -0.0010     0.1222  0.0637 0.1250 -0.0077 -0.0078
## 3   beta_X  0.3606  0.3324     0.0545  1.0000 0.0661 -0.4256 -0.5157
## 4 beta_XG2  0.1539  0.1404     0.1192  0.2370 0.1232 -0.1093 -0.1130
## 5 res.var  0.8400  0.7728     0.0901  0.6667 0.1175 -0.5721 -0.7457
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.007   1.156  1.011  60.047   1.001 218.432 294.843
##   neff_mean neff_q3 neff_max
## 1   318.987 421.398 602.985
##
## #####
## N = 500
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0177     0.0529  0.0608 0.0541 -0.1001 -0.1023
## 2   beta_G2 0.0000  0.0000     0.1092  0.0592 0.1100 -0.0004 -0.0004
## 3   beta_X  0.3606  0.3306     0.0490  0.9992 0.0620 -0.4835 -0.6118
## 4 beta_XG2  0.1539  0.1437     0.1063  0.2517 0.0970 -0.1051 -0.0958
## 5 res.var  0.8400  0.7723     0.0804  0.6667 0.1105 -0.6127 -0.8416
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true     est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0177     0.0529  0.0610 0.0541 -0.0993 -0.1015

```

```

## 2  beta_G2  0.0000 -0.0002      0.1092  0.0593 0.1101 -0.0018 -0.0018
## 3  beta_X   0.3606  0.3307      0.0487  1.0000 0.0614 -0.4859 -0.6128
## 4  beta_XG2 0.1539  0.1438      0.1063  0.2523 0.0970 -0.1033 -0.0943
## 5  res.var  0.8400  0.7723      0.0804  0.6667 0.1105 -0.6130 -0.8422
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.006    1.161   1.01  62.238    1.001 216.934  308.447
##   neff_mean neff_q3 neff_max
## 1   322.641 428.474  695.108
##
## #####
## N = 1000
## #####
## Crashed reps: 0 (0.0%)
## Successful reps: 200 (100.0%)
##
## -- Parameter Recovery (all successful reps) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0119    0.0374  0.0508 0.0353  0.0123  0.0116
## 2   beta_G2 0.0000  0.0040    0.0771  0.0650 0.0827  0.0484  0.0520
## 3   beta_X  0.3606  0.3356    0.0343  1.0000 0.0472 -0.5287 -0.7267
## 4  beta_XG2 0.1539  0.1408    0.0747  0.4700 0.0773 -0.1685 -0.1744
## 5  res.var  0.8400  0.7741    0.0567  0.6667 0.1000 -0.6594 -1.1618
##
## -- Parameter Recovery (excluding high PSR or low N_eff) --
##   param.id   true      est avg_post_sd pct_sig emp_sd bias_sd      z
## 1   beta_0 -0.0123 -0.0121    0.0374  0.0512 0.0353  0.0072  0.0068
## 2   beta_G2 0.0000  0.0040    0.0771  0.0655 0.0830  0.0477  0.0513
## 3   beta_X  0.3606  0.3355    0.0344  1.0000 0.0471 -0.5317 -0.7290
## 4  beta_XG2 0.1539  0.1412    0.0747  0.4710 0.0772 -0.1639 -0.1694
## 5  res.var  0.8400  0.7742    0.0568  0.6667 0.1000 -0.6580 -1.1597
##
## -- Convergence / N_eff Summary (all successful reps) --
##   psr_min psr_q1 psr_med psr_mean psr_q3 psr_max neff_min neff_q1 neff_med
## 1   1.001  1.004   1.006    1.008  1.011   1.048   64.315 235.113  309.932
##   neff_mean neff_q3 neff_max
## 1   334.325 445.471  653.26

```